Year 8

REVISION PACK

Unit 2

Name: _____

Class: _____

Unit 2 · Solving equations

Worked example: one-step and two-step equations

Solve

(a) $q + 6 = 15$ (b) $\frac{s}{5} = 4$ (c) $20 - 3y = 8$ (d) $\frac{2w}{3} = 10$

(a)	**Step 1**	Subtract 6	$q = 15 - 6$
	Step 2	Answer	$q = 9$
(b)	**Step 1**	Multiply by 5	$s = 4 \times 5$
	Step 2	Answer	$s = 20$
(c)	**Step 1**	Add $3y$, subtract 8	$20 - 8 = 3y$
	Step 2	Work out 20 − 8	$12 = 3y$
	Step 3	Answer	$y = 4$
(d)	**Step 1**	Multiply by 3	$2w = 30$
	Step 2	Divide by 2	$w = 15$

Practice: one-step and two-step equations

1.1 Solve

(a) $q + 5 = 13$ (b) $\frac{s}{4} = 6$ (c) $15 - 4y = 3$ (d) $\frac{3w}{4} = 18$

1.2 Solve

(a) $q + 7 = 19$ (b) $\frac{s}{3} = 6$ (c) $25 - 5y = 10$ (d) $\frac{5w}{2} = 15$

1.3 Solve

(a) $q + 9 = 14$ (b) $\frac{s}{6} = 2$ (c) $30 - 6y = 6$ (d) $\frac{3w}{5} = 9$

1.4 Solve

(a) $q + 12 = 30$ (b) $\frac{s}{2} = 9$ (c) $17 - 2y = 5$ (d) $\frac{4w}{3} = 16$

1.5 Solve

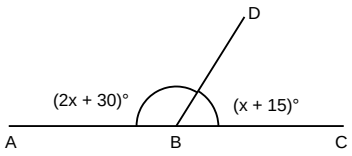
(a) $q + 3 = 21$ (b) $\frac{s}{8} = 5$ (c) $40 - 7y = 12$ (d) $\frac{2w}{7} = 6$

1.6 Solve

(a) $q + 15 = 27$ (b) $\frac{s}{7} = 4$ (c) $22 - 3y = 7$ (d) $\frac{7w}{2} = 14$

Unit 2 · Angles and equations

Worked example: angles on a straight line

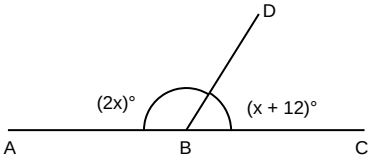


ABC is a straight line. Write an equation and solve it to find angle ABD.

Step 1	Angles on a line add to 180°	$(2x + 30) + (x + 15) = 180$
Step 2	Collect like terms	$3x + 45 = 180$
Step 3	Subtract 45	$3x = 135$
Step 4	Divide by 3	$x = 45$
Step 5	Angle ABD	$2 \times 45 + 30 = 120$
Step 6	Answer	120°

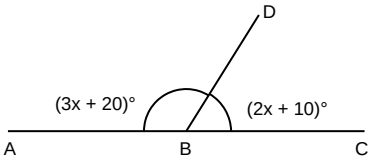
Practice: angles on a straight line

2.1



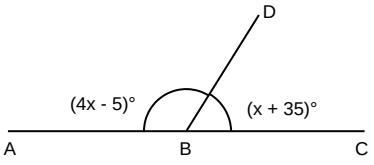
ABC is a straight line. Write an equation and solve it to find angle ABD.

2.2



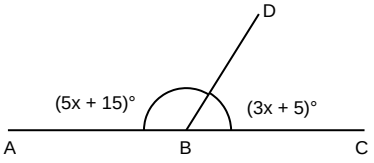
ABC is a straight line. Write an equation and solve it to find angle DBC.

2.3



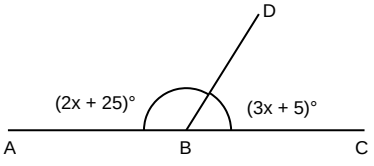
ABC is a straight line. Write an equation and solve it to find angle ABD.

2.4



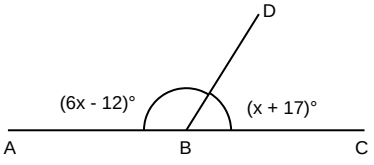
ABC is a straight line. Write an equation and solve it to find angle DBC.

2.5



ABC is a straight line. Write an equation and solve it to find angle ABD.

2.6



ABC is a straight line. Write an equation and solve it to find angle DBC.

Unit 2 · Forming equations

Worked example: writing and solving an equation from words

Dana is y years old. Omar is 3 years older than Dana. Ali is twice as old as Omar. The sum of their ages is 33 years.

(a) Write an equation for the sum of their ages.

(b) Solve the equation to find how old Ali is.

(a)	Step 1	Omar: $y + 3$. Ali: $2(y + 3)$	$y + (y + 3) + 2(y + 3) = 33$
	Step 2	Expand and simplify	$4y + 9 = 33$
(b)	Step 1	Subtract 9	$4y = 24$
	Step 2	Divide by 4	$y = 6$
	Step 3	Omar	$6 + 3 = 9$
	Step 4	Ali is twice Omar's age	$\text{Ali} = 2 \times 9 = 18$ years

Practice: writing and solving an equation from words

3.1 Ben is y years old. Liz is 2 years older than Ben. James is twice as old as Liz. The sum of their ages is 62 years.

(a) Write an equation for the sum of their ages.

(b) Solve the equation to find how old James is.

3.2 Sara is y years old. Noor is 4 years older than Sara. Adam is twice as old as Noor. The sum of their ages is 72 years.

(a) Write an equation for the sum of their ages.

(b) Solve the equation to find how old Adam is.

3.3 Hadi is y years old. Rana is 2 years younger than Hadi. Tariq is three times as old as Rana. The sum of their ages is 42 years.

(a) Write an equation for the sum of their ages.

(b) Solve the equation to find how old Tariq is.

3.4 Lina is y years old. Sami is 5 years older than Lina. Yusuf is three times as old as Sami. The sum of their ages is 75 years.

(a) Write an equation for the sum of their ages.

(b) Solve the equation to find how old Yusuf is.

3.5 Mona is y years old. Zaid is 3 years younger than Mona. Huda is twice as old as Zaid. The sum of their ages is 35 years.

(a) Write an equation for the sum of their ages.

(b) Solve the equation to find how old Huda is.

3.6 Faris is y years old. Dina is 6 years older than Faris. Karim is twice as old as Dina. The sum of their ages is 78 years.

(a) Write an equation for the sum of their ages.

(b) Solve the equation to find how old Karim is.

Unit 2 · Writing formulae

Worked example: writing and using a formula

Sam has \$$p$ in a bank account. Sam pays in a further \$20 every week for n weeks.

(a) Write a formula for the total T in the account after n weeks.

(b) One week the balance is \$150. Some weeks later it is \$270. How many weeks passed?

(a)	**Step 1**	Start + 20 each week	$T = p + 20n$
(b)	**Step 1**	How much more money	$270 - 150 = 120$
	Step 2	Divide by 20 a week	$120 \div 20 = 6$
	Step 3	Answer	6 weeks

Practice: writing and using a formula

4.1 Rahman has \$$p$ in a bank account. Rahman pays in a further \$50 every week for n weeks.

(a) Write a formula for the total T in the account after n weeks.

(b) One week the balance is \$524. Some weeks later it is \$974. How many weeks passed?

4.2 Aya has \$$p$ in a bank account. Aya pays in a further \$30 every week for n weeks.

(a) Write a formula for the total T in the account after n weeks.

(b) One week the balance is \$215. Some weeks later it is \$455. How many weeks passed?

4.3 Karim has \$$p$ in a bank account. Karim pays in a further \$25 every week for n weeks.

(a) Write a formula for the total T in the account after n weeks.

(b) One week the balance is \$340. Some weeks later it is \$640. How many weeks passed?

4.4 Laila has \$$p$ in a bank account. Laila pays in a further \$40 every week for n weeks.

(a) Write a formula for the total T in the account after n weeks.

(b) One week the balance is \$180. Some weeks later it is \$700. How many weeks passed?

4.5 Nabil has \$$p$ in a bank account. Nabil pays in a further \$15 every week for n weeks.

(a) Write a formula for the total T in the account after n weeks.

(b) One week the balance is \$92. Some weeks later it is \$257. How many weeks passed?

4.6 Salma has \$$p$ in a bank account. Salma pays in a further \$60 every week for n weeks.

(a) Write a formula for the total T in the account after n weeks.

(b) One week the balance is \$410. Some weeks later it is \$1010. How many weeks passed?

Unit 2 · Solving harder equations

Worked example: unknowns on both sides and brackets

Solve

(a) $5x + 2 = 2x + 14$ (b) $20 - 3y = 2(y + 5)$ (c) $3(t - 2) = t + 8$

(a)	**Step 1**	Subtract $2x$	$3x + 2 = 14$
	Step 2	Subtract 2	$3x = 12$
	Step 3	Answer	$x = 4$
(b)	**Step 1**	Expand	$20 - 3y = 2y + 10$
	Step 2	Add $3y$, subtract 10	$10 = 5y$
	Step 3	Answer	$y = 2$
(c)	**Step 1**	Expand	$3t - 6 = t + 8$
	Step 2	Subtract t, add 6	$2t = 14$
	Step 3	Answer	$t = 7$

Practice: unknowns on both sides and brackets

5.1 Solve

(a) $5x + 9 = 3x + 15$ (b) $34 - 4y = 5(y + 5)$ (c) $7(t - 4) = t + 2$

5.2 Solve

(a) $8x + 1 = 4x + 29$ (b) $20 - 2y = 3(y + 5)$ (c) $5(t - 3) = t + 9$

5.3 Solve

(a) $8x + 4 = 4x + 20$ (b) $32 - 2y = 4(y + 5)$ (c) $5(t - 2) = t + 22$

5.4 Solve

(a) $8x + 9 = 3x + 24$ (b) $10 - 2y = 4(y + 1)$ (c) $3(t - 2) = t + 6$

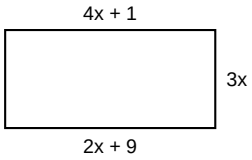
5.5 Solve

(a) $10x + 8 = 4x + 26$ (b) $38 - 2y = 4(y + 2)$ (c) $6(t - 2) = 3t + 15$

5.6 Solve

(a) $8x + 4 = 3x + 14$ (b) $10 - 4y = 2(y + 2)$ (c) $3(t - 4) = t + 2$

Worked example: equal sides of a rectangle

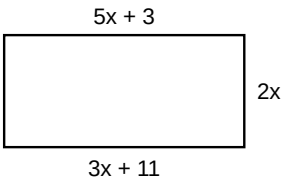


All the measurements of the rectangle are in cm.
(a) Write an equation for the two long sides.
(b) Find the perimeter of the rectangle.

(a)	Step 1	Opposite sides are equal	$4x + 1 = 2x + 9$
(b)	Step 1	Subtract $2x$, subtract 1	$2x = 8$
	Step 2	Divide by 2	$x = 4$
	Step 3	Length and width	$4 \times 4 + 1 = 17,\ \ 3 \times 4 = 12$
	Step 4	Add all four sides	$17 + 17 + 12 + 12 = 58\ \text{cm}$

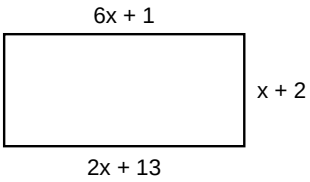
Practice: equal sides of a rectangle

6.1



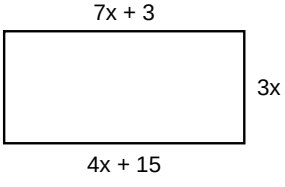
All the measurements of the rectangle are in cm.
(a) Write an equation for the two long sides.
(b) Find the perimeter of the rectangle.

6.2



All the measurements of the rectangle are in cm.
(a) Write an equation for the two long sides.
(b) Find the perimeter of the rectangle.

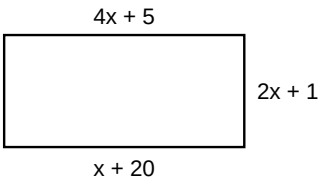
6.3



All the measurements of the rectangle are in cm.
(a) Write an equation for the two long sides.
(b) Find the perimeter of the rectangle.

Practice: equal sides of a rectangle (continued)

6.4

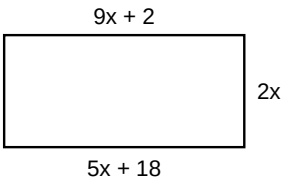


All the measurements of the rectangle are in cm.

(a) Write an equation for the two long sides.

(b) Find the perimeter of the rectangle.

6.5

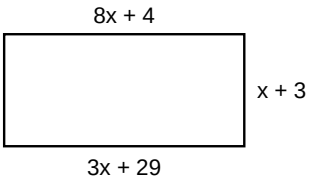


All the measurements of the rectangle are in cm.

(a) Write an equation for the two long sides.

(b) Find the perimeter of the rectangle.

6.6



All the measurements of the rectangle are in cm.

(a) Write an equation for the two long sides.

(b) Find the perimeter of the rectangle.